Design and Implementation of Real Time Controller for Self Balancing Robot

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***Abstract*—In this paper modeling and control of a two wheeled balancing autonomous robot is reported. The main focus is on developing efficient control algorithms required to enable the robot to perceive and act in real time for a dynamically changing world. The prototype relies on a multi-sensor system consisting of *i)* two optical incremental encoders and *ii)* a three axis accelerometer which is used as a tilt sensor. The information from the sensory system is provided to the robot controller. The performance and reliability of controller is then simulated in MATLAB® as a platform. Finally, the solution is verified on a real physical model controlled by means of a single chip AT89C52 microcontroller.**

***Keywords*— Simulation, dynamics, inverted pendulum mobile robot, state space control**

I. Introduction

Effective and efficient control system designs provide the robot with the ability to control itself and operate autonomously. Two wheeled robots are one variation of robot that has become a standard topic of research and exploration for young engineers and robotic enthusiasts. They offer the opportunity to develop control systems that are capable of maintaining stability of an otherwise unstable system. This type of system is also known as an inverted pendulum [1].

Two wheeled balancing robot is a classic engineering problem based on inverted pendulum and is much like trying to balance a broom on the tip of your finger. The word balance means the inverted pendulum is in equilibrium state, which its position is like standing upright 90 degrees. However, the system itself is not balance, which means it keeps falling off, away from the vertical axis. Therefore, a sensor is needed to provide the angle position of the inverted pendulum or robot base and input into the microcontroller, which the program in itself is a balancing algorithm. The microcontroller will then provide a type of feedback signal through PWM control to the H-bridge circuit to turn the motor clockwise or anticlockwise, thus balancing the robot [2].

This project implements a self-sustaining, two-wheeled, balancing robot capable of balancing on level surfaces. The robot maintains its balance by implementing a Proportional Integral Derivative (PID) algorithm, which determines the amount of wheel rotation required to keep the robot upright. The robot is able to remain balanced indefinitely regardless of small outside disturbances, and is able to move forwards, backwards, and rotate according to signals sent. The prototype used in this work relies on a dead-reckoning multi-sensor system, which will consist of i) two optical incremental encoders mounted on the motor shafts, to sense the relative angular positions and angular velocities of the wheels and ii) a 3-Axis accelerometer tilt sensor to measure the inclination and angular velocity. The code is written in C and compiled for the Atmel AT89C52 microcontroller, which is interfaced with the sensors and motors. The main goal of the microcontroller is to fuse the wheel encoder and accelerometer sensors to estimate the behaviour of the platform and then to use this information to drive the wheel in the direction to maintain an upright and balanced position.

II. Robot Design

The physical layout of the robot is shown in Fig 1 and the assembled robot is shown in Fig 2.. The robot's frame consists of two platforms that allow room for the placement of the electronic equipment. It is important to consider the overall weight of the robot because base and the wheels might bend outward due to the overall weight of the components. The entire component contributed in the overall weight is tabulated in the following Table 1.

TABLE 1
PARAMETER OF ROBOT

Item	Quantity	kg
Lead Acid batteries	3	1.19
Nylon material (sheets)	2	0.65
Printed circuit board	1	0.35
Motors, Bearings & couplings	2	1.5

Material that is used for building the chassis is polestar which is a strong, light and affordable. This setup is taken to ensure the portability of the robot [3].

III. System Modelling

One of the very first steps for a control system engineer to able to control a system successfully is to understand the system first. An engineer can understand the system better through modelling the system in a free body diagram. In this balancing wheeled robot the modelling is divided into two parts. The first is the wheel model and the second is the chassis model.

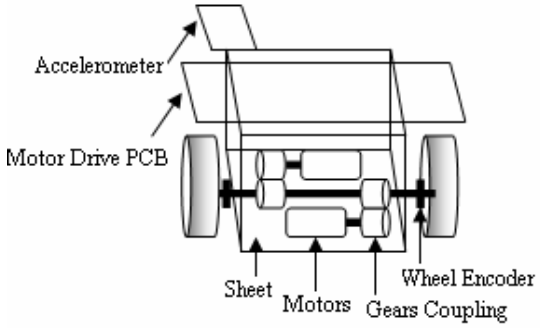


Fig.1 Physical Layout of the robot

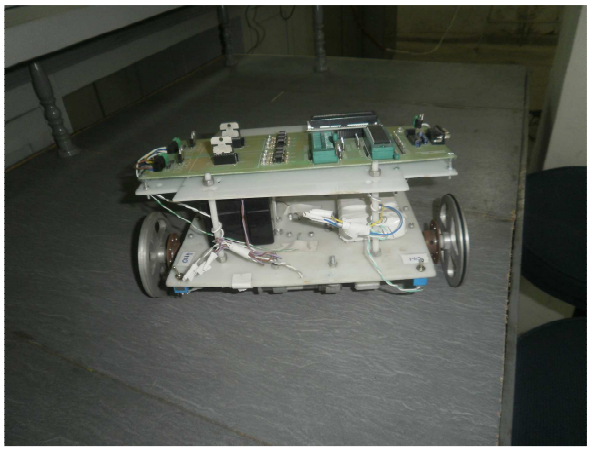


Fig.2 Assembled robot

A. Wheel Modelling

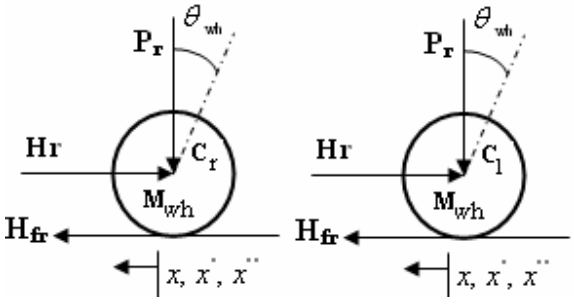


Fig.3 Free body diagram of the wheel

The forces acting on wheels are depicted in Fig 3. Using the Newton's law of motion, sum of the forces on the x-directions,

$$\sum F_x = Ma$$

$$H_{fr} - H_r = M_{wh}\ddot{x} \quad (1)$$

Summing up the moments around the centre of the wheel,

$$\sum M = J\alpha$$

$$C_r - H_{fr} \times r = J_{wh}\ddot{\theta} \quad (2)$$

Rearranging equation (2),

$$H_{fr} = \frac{C_r - J_{wh}\ddot{\theta}_{wh}}{r} \quad (3)$$

For the right wheel substitute equation (2) into (1),

$$M_{wh}\ddot{x} = \frac{C_r - J_{wh}\ddot{\theta}_{wh}}{r} - H_r \quad (4)$$

For the left wheel,

$$M_{wh}\ddot{x} = \frac{C_l - J_{wh}\ddot{\theta}_{wh}}{r} - H_l \quad (5)$$

Because the linear motion is acting on the centre of the wheel, angular motion can be transformed into linear motion by simple transformation.

$$\ddot{\theta}_{wh} \times r = \ddot{x}, \ \ddot{\theta}_{wh} = \frac{\ddot{x}}{r} \quad (6)$$

$$\dot{\theta}_{wh} \times r = \dot{x}, \ \dot{\theta}_{wh} = \frac{\dot{x}}{r} \quad (7)$$

For the right wheel,

$$M_{wh}\ddot{x} = \frac{C_r}{r} - \frac{J_{wh}}{r^2}\ddot{x} - H_r \quad (8)$$

For the left wheel,

$$M_{wh}\ddot{x} = \frac{C_l}{r} - \frac{J_{wh}}{r^2}\ddot{x} - H_l \quad (9)$$

Summing up equation (8) and (9),

$$2\left(M_{wh} + \frac{J_{wh}}{r^2}\right)\ddot{x} = \frac{C_l}{r} + \frac{C_r}{r} - (H_l + H_r) \quad (10)$$

B. Chassis Modelling

Chassis modelling parameters is shown in Fig 4. Sum of forces perpendicular to the chassis or pendulum,

$$\sum F_{perpendicular} = M_{rc}\ddot{x}\cos\theta_{rc}$$

$$(P_r + P_l)\sin\theta_{rc} + (H_r + H_l)\cos\theta_{rc} \quad (11)$$

$$-M_{rc}l\ddot{\theta}_{rc} - M_{rc}g\sin\theta_{rc} = M_{rc}\ddot{x}\cos\theta_{rc}$$

Sum of the forces in the horizontal direction

$$\sum F_{horizontal} = M_{rc}\ddot{x}$$

$$(H_r + H_l) + M_{rc}l\dot{\theta}^2_{rc}\sin\theta_{rc} - M_{rc}l\ddot{\theta}_{rc}\cos\theta_{rc} = M_{rc}\ddot{x} \quad (12)$$

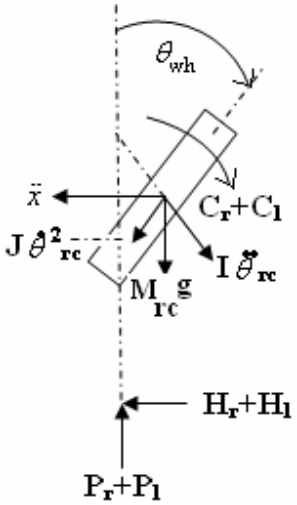


Fig.4 Inverted pendulum free body diagram

Sum of moments around the centre mass of the pendulum

$$\sum M=J\alpha \qquad \text{where } \alpha=\ddot{\theta}_{rc}$$

$$\left[-(P_r+P_l)l\sin\theta_{rc}-(H_r+H_l)l\cos\theta_{rc}-(C_r+C_l)=J_{rc}\ddot{\theta}_{rc}\right] \quad (13)$$

Rearranging equation (13),

$$-\left(P_r+P_l\right)l\sin\theta_{rc}-\left(H_r+H_l\right)l\cos\theta_{rc}=J_{rc}\ddot{\theta}_{rc}+\left(C_r+C_l\right) \quad (14)$$

Multiplying equation (11) by –ell,

$$\left[-(P_r+P_l)\sin\theta_{rc}-(H_r+H_l)\cos\theta_{rc}\right]l+M_{rc}gl\sin\theta_{rc}+M_{rc}l^2\ddot{\theta}_{rc}=-M_{rc}l\ddot{x}\cos\theta_{rc} \quad (15)$$

Substitute equation (14) into above equation to eliminate (P_r+P_l) and (H_r+H_l) term and rearranging the same term,

$$J_{rc}\ddot{\theta}_{rc}+\left(C_r+C_l\right)+M_{rc}gl\sin\theta_{rc}+M_{rc}l^2\ddot{\theta}_{rc}=-M_{rc}l\ddot{x}\cos\theta_{rc} \quad (16)$$

To eliminate (H_r+H_l) term from the second equation, equation (12) is inserted into equation (10) and rearranging the same terms,

$$2\left(M_{wh}+\frac{2J_{wh}}{r^2}+M_{wh}\right)\ddot{x}=\frac{C_l}{r}+\frac{C_r}{r}+M_{rc}l\ddot{\theta}_{rc}\cos\theta_{rc}+M_{rc}l\dot{\theta}^2_{rc}\sin\theta_{rc} \quad (17)$$

Equation (16) and (17) are non-linear equations, and to get linear equations, some linearising or assumption are taken. Let $\theta_{rc}=\pi+\delta$, where δ, is the small angle from the vertical upward direction. Therefore, $\cos\theta_{rc}=-1$ and $\frac{d^2\theta_{rc}}{dt^2}=0$.

The linearising equation representing the whole system is,

$$J_{rc}\ddot{\delta}_{rc}+\left(C_r+C_l\right)-M_{rc}gl\delta+M_{rc}l^2\ddot{\delta}_{rc} \quad (18)$$

$$2\left(M_{wh}+\frac{2J_{wh}}{r^2}+M_{rc}\right)\ddot{x}=\frac{C_l}{r}+\frac{C_r}{r}+M_{rc}l\ddot{\delta}_{rc} \quad (19)$$

Rearranging the state space representation,

$$\ddot{\delta}_{rc}=\frac{M_{rc}gl}{J_{rc}+M_{rc}l^2}\delta+\frac{M_{rc}l}{J_{rc}+M_{rc}l^2}\ddot{x}-\frac{\left(C_r+C_l\right)}{J_{rc}+M_{rc}l^2}$$

$$\ddot{x}=\frac{M_{rc}l}{\left(2M_{wh}+\frac{2J_{wh}}{r^2}+M_{rc}\right)}\ddot{\delta}+\frac{(C_r+C_l)}{r\left(2M_{wh}+\frac{2J_{wh}}{r^2}+M_{rc}\right)} \quad (20)$$

The state space equation is obtained as below after substituting equation (19) into (18) and equation (20) into (17) in the form of:

$$\frac{d\xi}{dt}=A\xi+Bu$$

$$\begin{bmatrix}\dot{x}\\ \ddot{x}\\ \dot{\delta}\\ \ddot{\delta}\end{bmatrix}=\begin{bmatrix}0 & 1 & 0 & 0\\ 0 & 0 & \frac{M^2_{rc}gl}{\phi-\frac{M^2_{rc}l}{J_{rc}+M_{rc}l^2}} & 0\\ 0 & 0 & 0 & 1\\ 0 & 0 & \frac{M^2_{rc}gl^2}{J_{rc}-\frac{M^2_{rc}l^2}{\phi}+M_{rc}l^2} & 0\end{bmatrix}\begin{bmatrix}x\\ \dot{x}\\ \dot{\delta}\\ \delta\end{bmatrix}+\begin{bmatrix}0 & 0\\ \frac{1}{r}-\frac{M_{rc}l}{J_{rc}+M_{rc}l^2} & \frac{1}{r}-\frac{M_{rc}l}{J_{rc}+M_{rc}l^2}\\ \frac{M_{rc}l}{r\phi}-1 & \frac{M_{rc}l}{r\phi}-1\end{bmatrix}\begin{bmatrix}C_r\\ C_l\end{bmatrix} \quad (21)$$

where $\phi=2M_{wh}+\frac{2J_{wh}}{r^2}+M_{rc}$ and output of $y=C\xi$

$$y = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} x \\ \dot{x} \\ \delta \\ \dot{\delta} \end{bmatrix} \quad (22)$$

The C matrix is oriented this way because the position of the wheel base and the robot chassis are interested.

Applying the feedback equation:

$$u = r - \begin{bmatrix} e & f & g & h \end{bmatrix} \begin{bmatrix} x1 \\ x2 \\ x3 \\ x4 \end{bmatrix} \quad (23)$$

Whereby x1, x2, x3, and x4 are the states of the state space equation.

The feedback equation is the equation that is about to inserted into the main system, with coefficient e, f, g and h are the arbitrary chosen values to be experimented (by trial and error) in order to get the robot to balance properly.

Finally the overall balancing robot control system is shown in the Fig 5 [3].

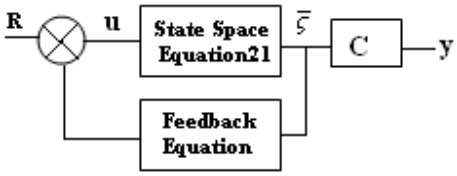


Fig.5 Balancing robot control system

The robot parameters are as tabulated in Table 2.

TABLE 2
PARAMETER OF ROBOT

Symbol	Parameter	Value [unit]
x	Linear Displacement	[m]
$\dot{x}$	Linear velocity	[m/sec]
$\ddot{x}$	Linear acceleration	[m/sec^2]
θ_{rc}	Rotational angle of robot chassis	[rad]
$\dot{\theta}_{rc}$	Angular velocity of robot chassis	[m/sec]
$\ddot{\theta}_{rc}$	Angular acceleration of robot chassis	[m/sec^2]
$\dot{\theta}_{wh}$	Angular velocity of the wheel	[m/sec]
$\ddot{\theta}_{wh}$	Angular acceleration of the wheel	[m/sec^2]
C_l	Applied torque from motor to the left wheel	
C_r	Applied torque from motor to the right wheel	
P_l, P_r	Reaction forces between the wheel and chassis	
H_l, H_r		
H_{fl}, H_{fr}	Friction forces between the wheels and the ground	
g	Gravitational constant	9.8 [m/s^2]
J_{rc}	Moment of inertia of robot chassis	[0.0458kg/m^2]
J_{wh}	Moment of inertia of robot wheels	[0.00020365kg/m^2]
M_{wh}	Mass of the robot wheels	[0.145kg]
M_{rc}	Mass of the robot chassis	[3.694kg]
l	Distance between the centre of the wheels and robot center of gravity	[0.165m]
r	Radius of the wheel	[0.053m]

Putting the above calculated parameter in equation 21 and the updated matrix is as follow

$$\begin{bmatrix} \dot{x} \\ \ddot{x} \\ \dot{\delta} \\ \ddot{\delta} \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 2.27 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 65.61 & 0 \end{bmatrix} \begin{bmatrix} x \\ \dot{x} \\ \delta \\ \dot{\delta} \end{bmatrix} + \begin{bmatrix} 0 & 0 \\ 14.68 & 14.68 \\ 1.786 & 1.786 \end{bmatrix} \begin{bmatrix} C_r \\ C_l \end{bmatrix} \quad (24)$$

IV. CONTROLLER DESIGNING

After conducting the linearization, the mathematical model is applied to develop the control scheme.

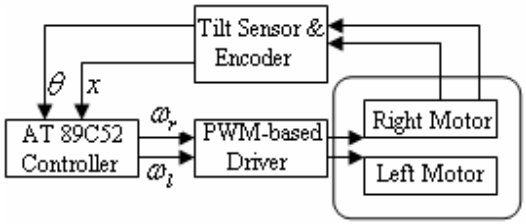


Fig.6 Block diagram of the entire model control system

As shown in the Fig 6, the tilt sensor and encoders measured four variables. All the variables is feedback to the controller. The controller computes the state space variables and produces the control inputs to stabilize the robot. Thus the controller computes the voltage which is then decoupled and modified to the actual voltage to be applied to the right and left wheels [4].

A. PID Controller Design

Proportional –integral-derivative (PID) control and the phase lead-lag compensation have been not only widely used in industrial control systems for achieving setpoint (or trajectory) tracking and regulation, but also extensively taught in feedback control courses [5], [6], due to their simple structure, robustness against plant variation, and easy parameter tuning. PID control generates control signals that are proportional to the error between the reference signal and the actual output (proportional action), to the integral of the error (integral action), thereby providing the desired control performance if the three-term parameters (proportional, integral and derivative gains) are well designed or tuned. Many classical and advance methods [7]-[9] have been proposed to design or tune these parameters in order to achieve the desired transient and steady-state performance specification.

The first thing is to do when using PID control in Matlab® is to find the transfer function of the system. The interest is lying in to control the robot position, which should return to the vertical after the initial disturbance, the reference signal should be zero. The force applied by the motors can be added as an impulse disturbance [10]. The closed loop transfer function is designed to make the settling time less than 2 and steady state error to the input reference should be zero [11]. The schematic of this problem is shown in Fig 7.

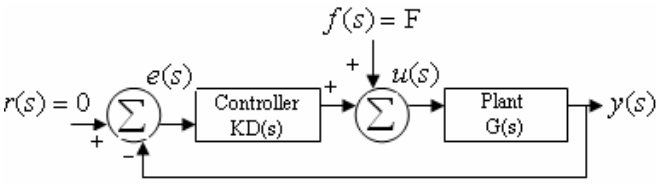


Fig.7 Block diagram of PID control system

The closed loop transfer function is modeled in Matlab and the impulse response plot with PID controller is shown in Fig 9. The response is still not stable. Now by increasing the proportional control to the system, increase the k to note the effect on the response as shown in Fig 10. The settling time is acceptable at about 2 seconds. By increasing the derivative control, satisfactory results are obtained as shown in Fig 11.

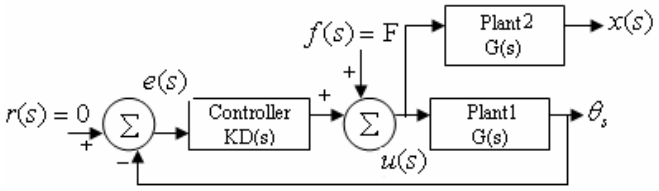


Fig.8 Block diagram of PID self balancing control system

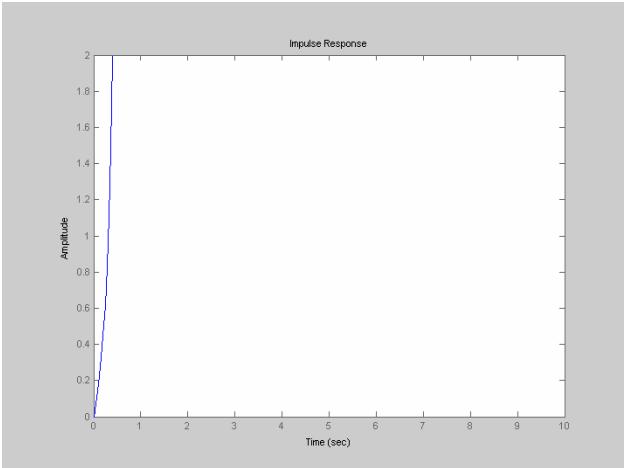


Fig.9 Impulse response with PID controller k=1, kd=1, ki=1

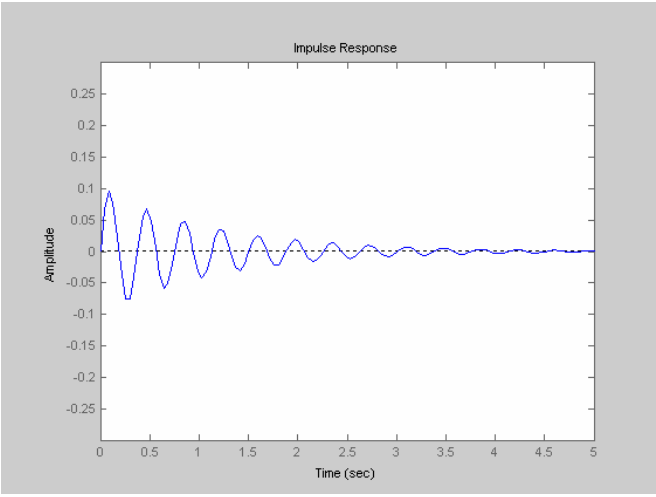


Fig.10 Impulse response with PID controller k=100, kd=1, ki=1

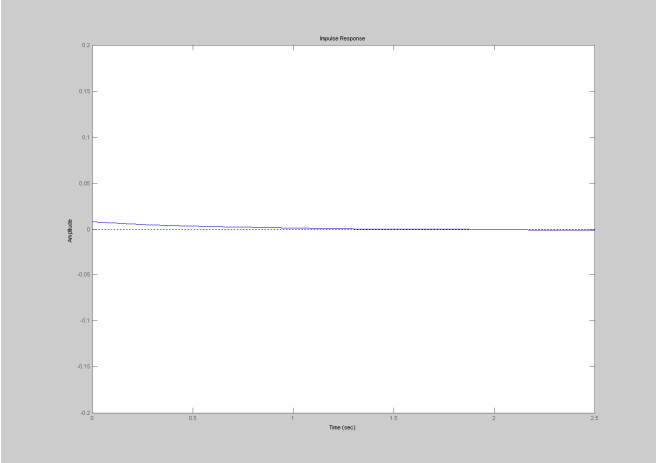


Fig.11 Impulse response with PID control k=100, kd=80, ki=45

The complete block diagram of self balancing control system is shown in Fig. 8. It is interesting to note what happen to the robot position when the controller for the robot's angle is in place. The impulse response of the robot position with PID control is shown in Fig 12. This shows that the robot moves in the negative direction with a variable velocity to stabilize itself.

The root locus sketch with k=100, kd=80, ki=45 is shown in Fig. 13. The PID controller is implemented in real time by using the discrete time representation with sampling time T_s 50msec [1].

$$\int_0^t error(t)dt \approx T_S \sum_{k=1} error(k)$$

$$\frac{derror(t)}{dt} \approx \frac{error(k) - error(k-1)}{T_S}$$

$$u(k) \approx K_p \left(error(k) + \frac{T_S}{T_I} \sum_{k=1} error(k) + \frac{T_D}{T_S} (error(k) - error(k-1)) \right)$$

where

$$K_I = \frac{T_p T_s}{T_I}$$

$$K_D = \frac{K_p T_D}{T_s}$$

$$u(k) = K_p error(k) + K_I \sum_{K=1} error(k) + K_D (error(k) - error(k-1))$$

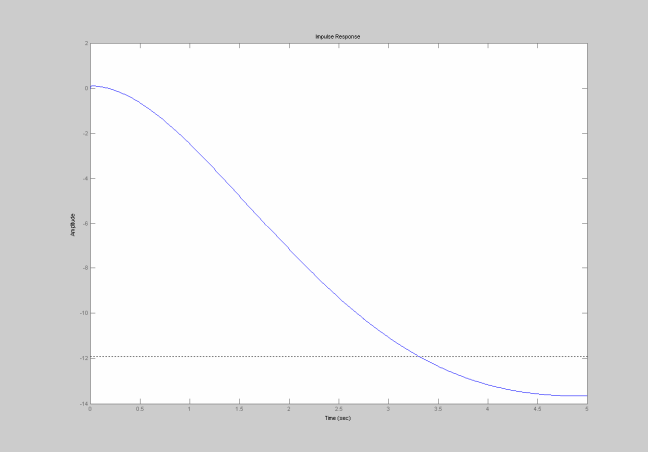


Fig.12 Impulse response of the robot position with PID control

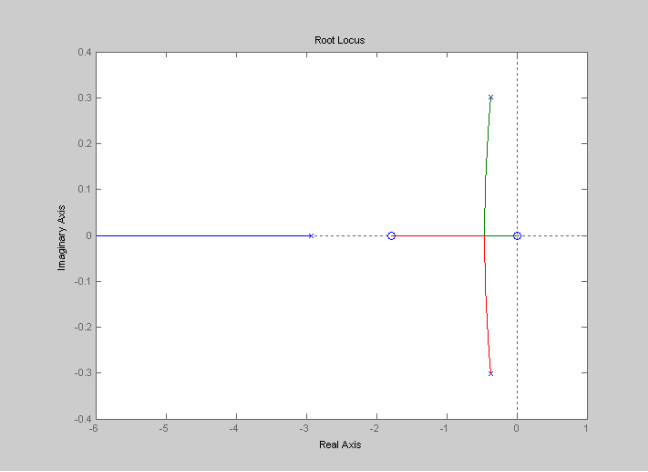


Fig.13 Root Locus sketch with PID control k=100, kd=80, ki=45

V. Hardware Design

The hardware of the robot consists of motor interface board, accelerometer and microcontroller board

A. *Motor Interface Board*

The motor interface circuit contains two L298 H-bridges each of which is connected in a parallel manner to enhance the current rating. Blocking diodes is used to block the reverse current or the leakage current. 4N25 optocouplers is being used to provide the isolation between the controller side and to the motor drive section. The circuit uses three inputs from the controller.

The motor interface circuit is shown in Fig 15 and microcontroller along with motor interface circuit is shown in Fig 16.

TABLE 3
Motor Interface Circuit Operation

Inputs Status			Motor Status
IN1	IN2	En	
0	0	0	Stop
0	1	1	Forward
1	0	1	Backward
1	1	1	Stop

B. *Accelerometer*

The Hitachi H48C Tri-Axis Accelerometer is an integrated module that can sense gravitational (g) force of ±3g on three axes (X, Y, and Z). The module contains an onboard regulator to provide 3.3-volt power to the H48C, analog signal conditioning, and an MCP3204 (four channel, 12-bit) analog-to-digital converter to read the H48C voltage outputs. All components are mounted on a 0.7 by 0.8 inch module. Acquiring measurements from the module is simplified through a synchronous serial interface. With the BASIC Stamp® series, for example, this is easily handled with the SHIFTOUT and SHIFTIN commands. Features: Measure ±3 g on any axis Uses MEMS (Micro Electro-Mechanical System) technology Onboard regulator and high-resolution ADC for simple connection to microcontroller host SPI communications compatible with BASIC Stamp 2 series SHIFTOUT and SHIFTIN commands Free-fall output indicates simultaneous 0g an all axes Small, breadboard-friendly package Key Specifications: Power Requirements: 5 VDC Communication: Serial SPI Dimensions: 0.7 x 0.8 x .45 in (17.8 x 20.3 x 11.4 mm) Operating Temperature: -13 to +167 °F (-25 to +75 °C) [12].

The accelerometer board is shown in Fig. 14.

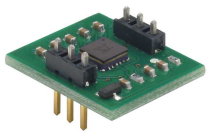


Fig. 14 Accelerometer Board

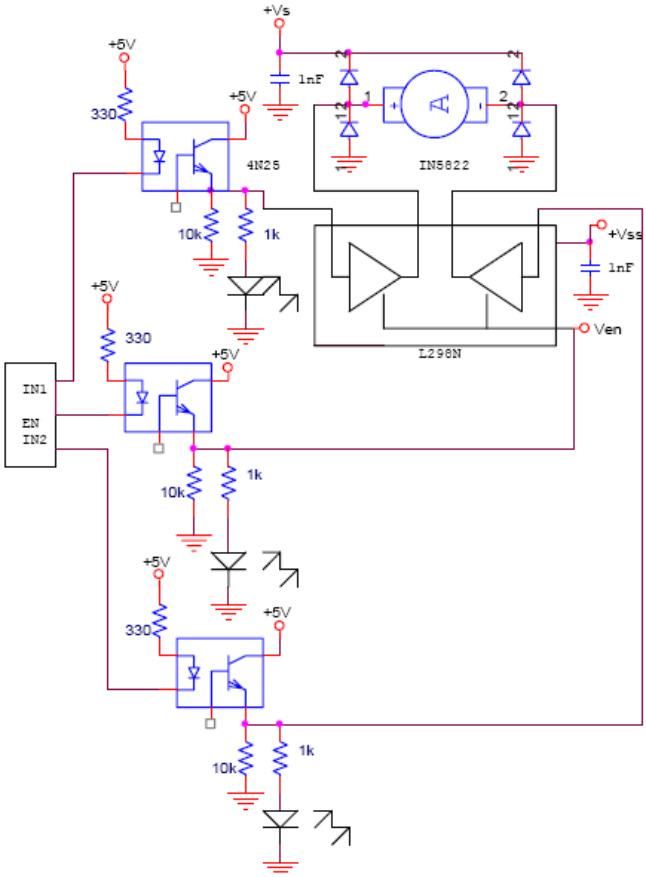


Fig. 15 Motor interface circuit

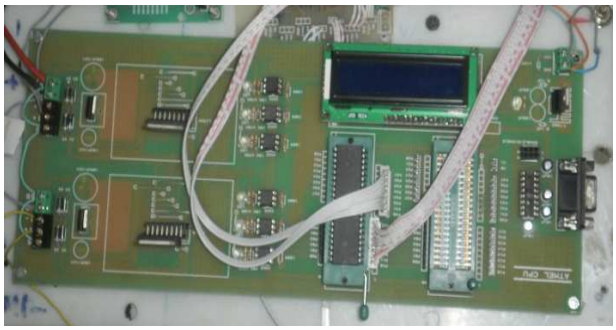


Fig. 16 Motor Controller Board

VI. Conclusion

In this paper, an effective and pragmatic approach to the modeling and control of two wheeled balancing robot using low cost components is presented. Sensory system of the robot consists of accelerometer and encoders which gives the tilt and the displacement information respectively. A linear PID controller has been developed that helps the robot to balance itself properly. The implementation of the robots hardware was easily achieved providing the system integration and functionality. Being based on low cost components, the project is an ideal demonstration of control theory for a classroom.

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References

[1] Mr Peter Miler, *"Buliding a Two Wheeled Balancing Robot"*, University of Southern Queensland, October, 2008.

[2] Soumit Kumar Biswal, "Development of Self Balanced Robot and its Controller", *National Institute of Technology, Rourkela, 2009*.

[3] Ho-Khoon Chye Randel, *"Balancing Wheel Robot"*, University of Southern Queensland, October 2005.

[4] M.H. Knudsen, "Experimental modeling of dynamic systems: An educational approach", *IEEE Trans. Educ*., vol. 41, pp. 146-158, May 1998.

[5] R. Kelly and J. Moreno, "Learning PID structures in an introductory courses of automatic control", *IEEE Trans. Educ*., vol. 44, pp. 373-376, Nov. 2001.

[6] J.C. Basilio and S.R. Matos, "Design of PI and PID controllers with transient performance specification", *IEEE Tans. Educ*., vol. 45, pp. 364-370, Nov. 2002.

[7] R.C. Dorf and R.H. Bishop, *"Modern Control Systems"*, 10th ed. Englewood Cliffs, NJ: Prentice-Hall, 2005.

[8] B.C. Kuo and F. Golnaraghi, *"Automatic control System"*, 8th ed. New York: Wiley, 2003.

[9] G.F. franklin, J. D. Powell, and A. Emami-Naeini, *"Feedback Control of Dynamic Systems"*, 4th ed. Englewood Cliffs, NJ: Prentice-Hall, 2002.

[10] Shui-Chun Lin and Ching-Chih Tsai, "Development of a self-balancing human transportation vehicle for the teaching of feedback control", *IEEE Trans. Educ*, vol. 52, No. 1, February 2009.

[11] Nawawi S. W, Ahmad. M. N, Osman J.H.S, Husain A. R and Abdollah A.F, "Controller design for two-wheels inverted pendulum mobile robot using PISMC", *4th Student Conference on Research and Development* (SCOReD 2006), Shah Alam, Selangor, Malaysia 27-28 June, 2006.

[12] [Online] visit: Toptronic (PTY) Ltd T A AP ELECTRONICS - HITACHI H48C TRI-AXIS ACCELEROMETER MODULE.